

# PING-AI state comparison

nb050 · 9 June 2026

## Abstract

One network (COBANet) can run as the gamma rhythm **PING** or as the Vreeswijk-Sompolinsky **asynchronous-irregular** state. With everything else harmonised — connectivity, weights,  $W^{II}$  — the two differ in a single control: the **correlation of the input** across cells. Shared (correlated) drive gives gamma; per-cell independent drive gives the balanced state.

## Methods

### Model parameters

The two models are identical COBANet except for the input. The table is the full configuration; the only rows that differ — all of them describing the input — are in *italics*.

| Parameter                       | PING                                                               | V&S AI                                                                  |
|---------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------|
| Model                           | COBANet (conductance LIF)                                          | COBANet (conductance LIF)                                               |
| $N_E / N_I$                     | 1024 / 256                                                         | 1024 / 256                                                              |
| $\Delta t / T$                  | 0.25 ms / 1000 ms                                                  | 0.25 ms / 1000 ms                                                       |
| $\tau_E / \tau_I$ (AMPA / GABA) | 2 / 9 ms                                                           | 2 / 9 ms                                                                |
| Recurrent connectivity          | sparse fixed-fan-in, $p = 0.2$                                     | sparse fixed-fan-in, $p = 0.2$                                          |
| $W^{EI}$ (E→I)                  | $\mathcal{N}(0.6, 0.18)$ $\mu\text{S}$                             | $\mathcal{N}(0.6, 0.18)$ $\mu\text{S}$                                  |
| $W^{IE}$ (I→E)                  | $\mathcal{N}(3.0, 0.9)$ $\mu\text{S}$                              | $\mathcal{N}(3.0, 0.9)$ $\mu\text{S}$                                   |
| $W^{II}$ (I→I)                  | none                                                               | none                                                                    |
| External drive                  | <i>dense <math>W_{\text{in}}</math> (shared 1024-neuron layer)</i> | <i>diagonal <math>W_{\text{in}}</math> (one private input per cell)</i> |
| Drive rate                      | <i>45 Hz (E)</i>                                                   | <i>45 Hz (E), 8 Hz (I)</i>                                              |
| Input correlation across cells  | <i>correlated</i>                                                  | <i>independent</i>                                                      |

Both models use sparse fixed-fan-in recurrent connectivity at  $p = 1 - s = 0.2$  on the  $E \leftrightarrow I$  matrices ( $W^{EE} = 0$ ,  $W^{II}$  off). Because  $p$  is also the fraction of recurrent input any two cells share ( $\langle n_{\text{shared}} \rangle / K = K / N = p$ ), it sets how strongly the loop can cohere — connectivity is the *internal* counterpart of the input knob. We fix it at a cortically realistic  $p = 0.2$  (local  $E \leftrightarrow I$  connectivity runs  $\approx 0.1$ – $0.5$ ) — dense enough for the recurrent loop to cohere, so connectivity is shared and does not distinguish the regimes.

### Model differences

The models share everything above — connectivity, the  $E \leftrightarrow I$  weights,  $W^{II}$ . The single difference is the **input**: a shared (correlated) drive for PING, per-cell independent (uncorrelated) drive for the V&S cell.

Both deliver 45 Hz Poisson through the *same* input pathway — spikes routed by  $W_{\text{in}}$  into a decaying AMPA conductance. The only difference is the **structure of  $W_{\text{in}}$** . The external excitatory conductance onto E cell  $j$  obeys

$$\tau_E \dot{g}_j^{\text{ext}}(t) = -g_j^{\text{ext}}(t) + \sum_k W_{jk}^{\text{in}} s_k(t), \quad W^{\text{in}} = \begin{cases} \text{dense} & (\text{PING}) \\ g_0 \mathbf{1} \text{ (diagonal)} & (\text{V\&S AI}) \end{cases}$$

where:

- $g_j^{\text{ext}}(t)$  — external excitatory conductance onto E cell  $j$ ;
- $\tau_E$  — AMPA decay time constant (2 ms), common to both;
- $s_k(t) = \sum_m \delta(t - t_k^m)$  — spike train of input neuron  $k$  (Poisson, 45 Hz);  $N_{\text{in}} = 1024$ ;
- $W^{\text{in}}$  — input weight matrix: **dense** for PING (every E cell reads the whole input layer, so all cells share the same drive) or **diagonal**  $g_0 \mathbf{1}$  for the V&S cell (each E cell reads only its own input neuron,  $W_{jk}^{\text{in}} = g_0 \delta_{jk}$ , so the drives are independent);
- $g_0$  — per-spike conductance, 0.38  $\mu\text{S}$  (the diagonal weight).

A dense  $W_{\text{in}}$  correlates the input across cells; a diagonal  $W_{\text{in}}$  leaves it independent. (In the runner the diagonal case is realized as a per-cell independent conductance rather than a literal identity matrix — the two are equivalent.)

The consequence is the cross-cell correlation of the drive:

$$\text{Corr}(g_j^{\text{ext}}, g_{j'}^{\text{ext}}) > 0 \quad (\text{PING, shared layer}), \quad \text{Corr}(g_j^{\text{ext}}, g_{j'}^{\text{ext}}) = 0 \quad (\text{V\&S AI, private streams}), \quad j \neq j'.$$

The shared layer is a *common reference* the loop can lock onto; the private streams kick each cell independently, with no common phase. Both drive E at the same 45 Hz, so rate is not the difference. What both share: neither input carries any rhythm. Both are featureless, steady Poisson (Figure 1), so PING's gamma is generated by the network, not inherited from the drive.

## Results

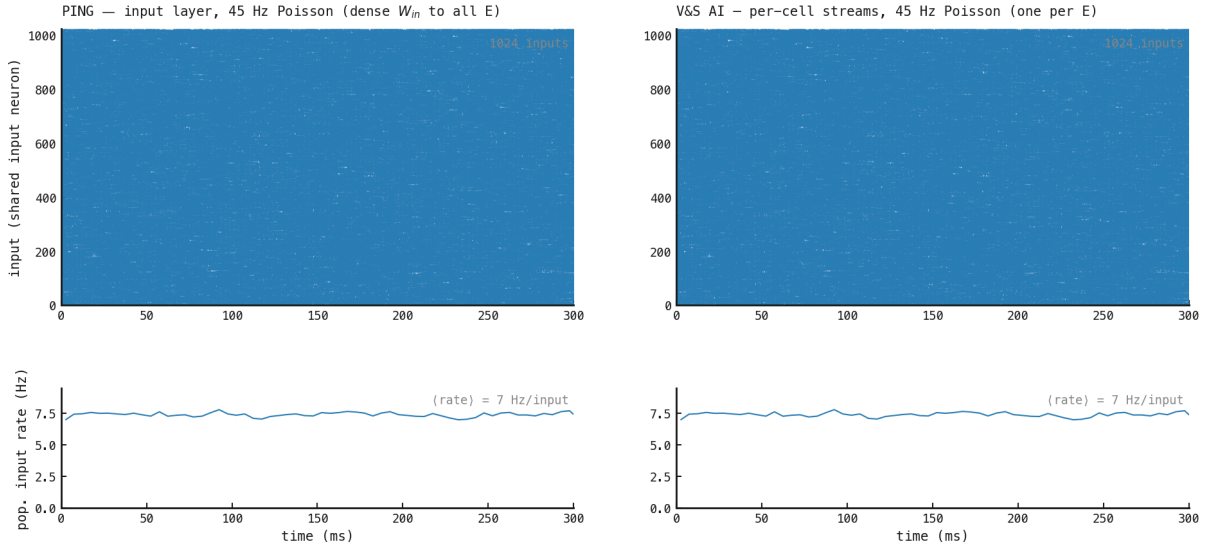


Figure 155: Top: a 300 ms raster of the drive — PING's shared input layer (left) and the V&S cell's per-cell private streams (right), both 45 Hz. Bottom: the population rate of each drive. Both are featureless, steady Poisson at the same rate; the input carries no oscillation in either regime. The regimes differ only in *correlation structure* — shared layer vs private streams — not in rate or temporal content.

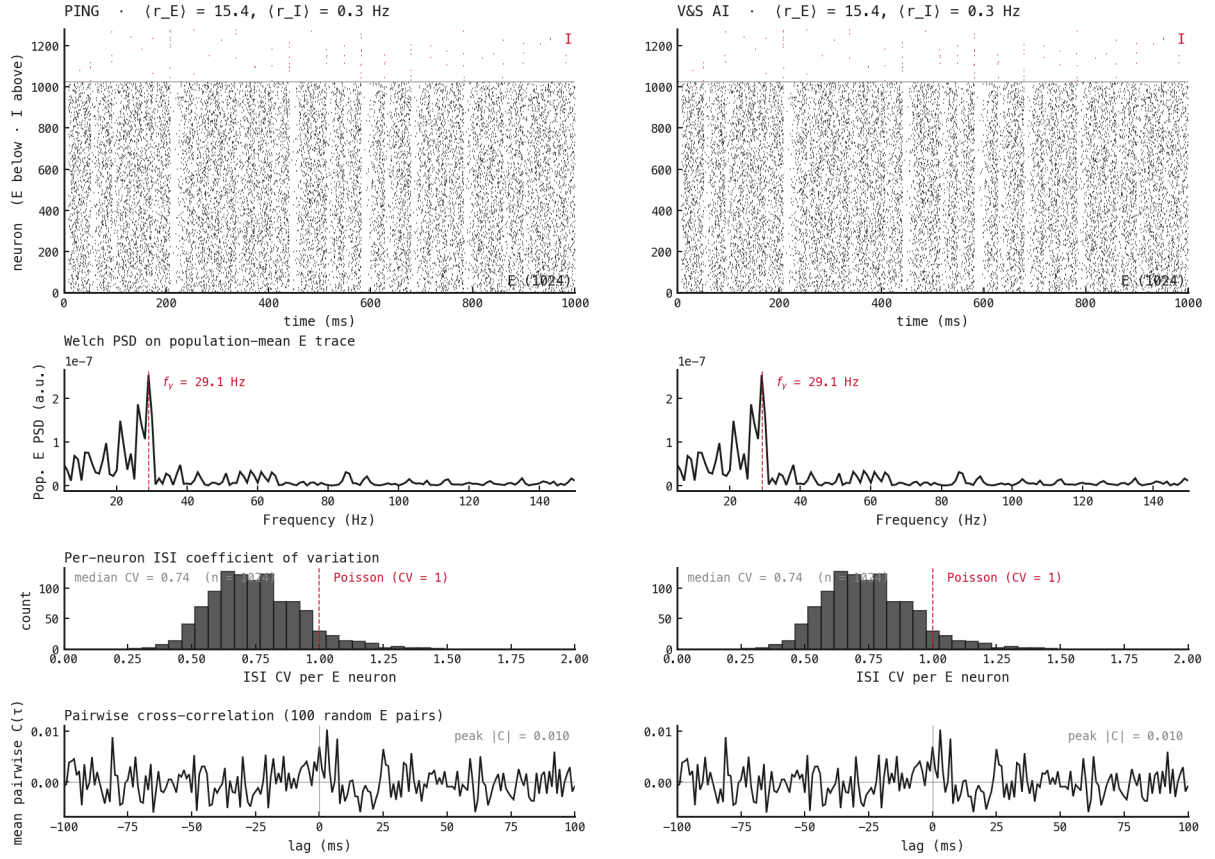


Figure 156: Row 1 is the raster with both populations on one axis — E (black) below, I (red) above the divider — so the I-leads-E gamma timing is visible directly. Both networks have identical wiring (sparse  $p = 0.2$ , matched weights, no  $W^{II}$ ); only the input differs. Left, PING (shared input): cycle-locked gamma bursts, a sharp 32 Hz peak, regular (sub-Poisson) ISIs (CV 0.64), cross-correlations combing at the gamma period (peak 0.063). Right, V&S AI (independent input): scattered raster, broadband spectrum, irregular ISIs, flat cross-correlation (peak 0.010). Same network and wiring, opposite dynamics — the input correlation alone.

## Next steps

The two regimes are now identical except for the input — same connectivity, weights, and 45 Hz E-drive rate — and the difference is its **cross-cell correlation**: a shared input layer (PING) versus per-cell independent streams (V&S). (One residual asymmetry: the V&S cell drives its I pool with an independent 8 Hz stream to decorrelate it, where PING’s I is loop-driven.) Turning correlation into a single, continuous control is the remaining work.

**A faithful correlation knob.** The tidy idea — mix a shared common stream with private streams and dial the shared fraction  $c \in [0, 1]$  — breaks at the  $c = 1$  extreme if “shared” is *one* stream broadcast verbatim: every cell then receives identical input and locks into trivial synchrony (cross-correlation  $\approx 0.99$ , a broad  $\approx 74$  Hz peak) rather than loop-generated gamma. Real correlated drive is a shared *population*, not one identical signal. The fix is a shared *layer* of  $K_{\text{shared}}$  common input neurons (each cell reading all of them), interpolated against private streams at matched rate — correlated without being identical, so genuine gamma survives. With that knob, the gamma $\leftrightarrow$ asynchronous boundary can be mapped continuously over  $c$ .